Yoav Gudes

Semiconductor Engineering Co-op Student
University of Waterloo 2028
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Summary

- Waterloo Nanotechnology Engineering student (Nanoelectronics & Quantum Engineering) with 16 months of co-op.
- Strong theoretical foundation in semiconductor physics and devices, microfabrication, thin films and quantum mechanics.
- Hands-on experience in cleanroom fabrication, nanomaterial synthesis, device characterization and circuit simulation.

Education

Nanotechnology Engineering: Nanoelectronics & Quantum Engineering Options | University of Waterloo **Sept 2023 – Apr 2028**
Relevant Coursework: Semiconductor Physics & Devices, Microfabrication & Thin-Film Technology, Quantum Mechanics

Lab Projects

- **MIS Capacitor Fabrication:** Built MIS capacitors on a 3-inch Si wafer via PECVD, lithography, RIE and sputtering.
- **Quantum Dot LED:** Synthesized CdSe quantum dots at six sizes and built a QD LED with size-tuned bandgap emission.
- **MOSFET Amplifier Design:** Designed and biased a common-source amplifier in LTspice for 10 V/V small-signal gain.

Work Experience

Process Engineering Intern | WOLF Advanced Technology **Sept 2025 – Apr 2026**

- Led stack-up failure analysis across 10 failed GPU units, tracing root causes to assembly and design issues.
- Validated 50+ NVIDIA/AMD GPU modules through −45 °C to +85 °C thermal cycling, logging test data with Python scripts.
- Designed heat pipe and vapor chamber assemblies in SolidWorks, prototyping a higher-conductivity variant via cold spray.

AI/ML Engineering & Data Science Intern | TD Bank **Jan 2025 – Apr 2025**

- Developed an NLP sentiment classifier on 10,000+ labeled utterances, presented to senior staff as an Azure AI alternative.

Data Engineering Intern | MedReddie **May 2024 – Aug 2024**

- Built a Python and LLM API pipeline to collect, validate and de-duplicate data for 8,000+ suppliers and $50M+ products.

Technical Skills

Semiconductor Fabrication & Processing

- *Deposition:* PECVD & DC Sputtering
- *Photolithography:* Multi-Level Patterning & UV Exposure
- *Etching:* Reactive Ion Etching & Wet Chemical Etching
- *Solution Processing:* Spin Coating & Vacuum Baking

Materials & Electrical Characterization

- *Electrical:* C–V/I–V, 4-Point Probe & Probe Station
- *Microscopy:* SEM, TEM, AFM & Optical Microscopy
- *Spectroscopy:* FTIR, Raman & UV–Vis
- *Thin Films:* Ellipsometry, Profilometry & XRD

Semiconductor Devices & Circuits

- *Devices:* Diodes, MIS Capacitors, MOSFETs & CMOS Logic
- *Analog Circuits:* Biasing, Small-Signal Analysis & Amplifiers
- *Optoelectronics:* LEDs, Lasers & Photodetectors

Software & Data

- *Programming:* Python, MATLAB & SQL
- *CAD & Simulation:* SolidWorks & LTspice
- *Data Analysis:* Python Scripting & Test Data Logging

Leadership & Achievements

Student Body President | Hillel Waterloo & Laurier **May 2025 – Aug 2026**

Awards

- **University of Waterloo:** Term Distinction (Dean's List), Excellent Academic Standing & Co-op Review, Jordan Banks Award.
- **TD Bank:** Achieve the Dream Canada Scholarship, Co-op Challenge Finalist

Languages: English & Hebrew (Fluent); Portuguese & Spanish (Conversational)